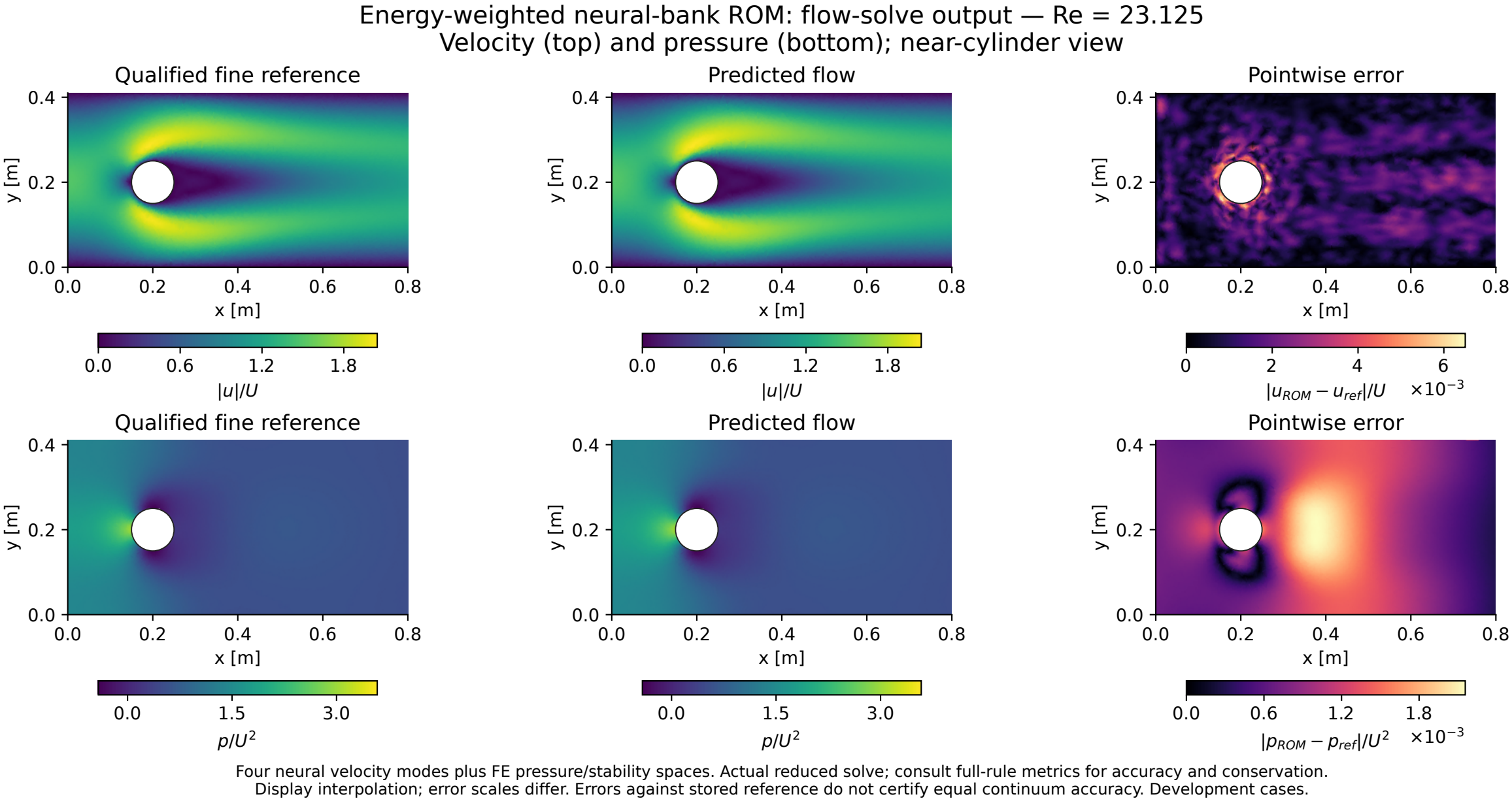

Energy-weighted neural-bank ROM: flow-solve output — Re = 23.125
Velocity (top) and pressure (bottom); near-cylinder view
Qualified fine reference
Predicted flow
Pointwise error
y [m]
x [m]
|u|/U
|u_ROM − u_ref|/U
×10^−3
Qualified fine reference
Predicted flow
Pointwise error
p/U²
|p_ROM − p_ref|/U²
×10^−3
Four neural velocity modes plus FE pressure/stability spaces. Actual reduced solve; consult full-rule metrics for accuracy and conservation.
Display interpolation; error scales differ. Errors against stored reference do not certify equal continuum accuracy. Development cases.